SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

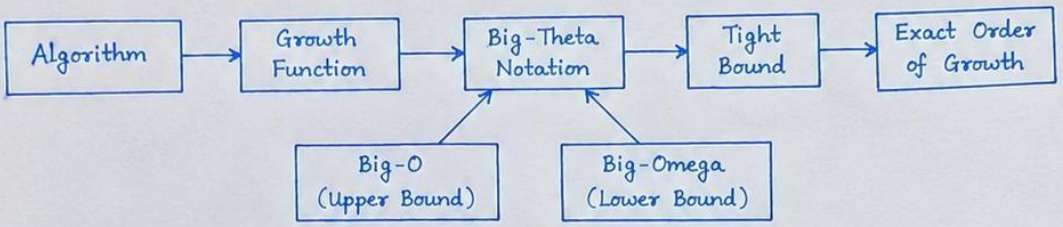
RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

8. Consider the concept map: Algorithm → Growth Function → Big-Theta Notation → Tight Bound → Exact Order of Growth. Explain how the concept of tight bound emerges when upper and lower bounds meet. Extend the map by adding links from Big-O and Big-Omega toward Big-Theta to show their relationship. Justify why Big-Theta is often preferred when the exact asymptotic growth is known. Interpret how the map helps distinguish between approximate and precise complexity descriptions.

Answer:

The completed concept map is shown below:



1. **Algorithm → Growth Function:**
   - The algorithm's operations are analyzed to form a growth function $f(n)$.
   - The growth function shows how the running time increases with input size.

2. **Growth Function → Big-O, Big-Omega, Big-Theta:**
   - Big-O gives an upper bound on the growth rate.
   - Big-Omega gives a lower bound on the growth rate.
   - Big-Theta gives a tight bound where upper and lower bounds meet.

3. **Big-O and Big-Omega → Big-Theta:**
   - When the upper and lower bounds are of the same order (i.e., $f(n) = \Theta(g(n))$), Big-O, Big-Omega and Big-Theta converge.
   - Thus, Big-Theta represents the exact order of growth.

4. **Big-Theta → Tight Bound → Exact Order of Growth:**
   - Big-Theta provides a tight bound, meaning the function grows neither faster nor slower than the given rate.
   - This leads to the exact order of growth (e.g., $\Theta(n \log n)$, $\Theta(n^2)$, etc.).

5. **Why Big-Theta is Preferred:**
   - It gives the precise asymptotic growth when both upper and lower bounds are known.
   - It is more informative and useful for algorithm comparison and analysis.

6. **Interpretation:**

   This concept map shows that Big-O and Big-Omega give approximate bounds, while Big-Theta gives the exact order when they meet. It helps distinguish between approximate complexity descriptions (using O or Ω) and precise descriptions (using Θ). Hence, Big-Theta is preferred when the exact asymptotic growth is known.